Virtue of Complexity or Volatility-Timed Momentum? A Placebo Test of Over-Parameterized Market Timing

Lifantao (Teddy) He
University of Pennsylvania
teddyhe@engineering.upenn.edu

Abstract

Kelly, Malamud and Zhou argue that ridge regression on thousands of random features of about fifteen predictors times the U.S. stock market better than simple models, even when trained on twelve months of data. Nagel argues that such a model does not learn from the predictors: with a short window it reduces mechanically to volatility-timed momentum. We test the two explanations on monthly data from 1927 to 2025 with a strictly walk-forward design, twenty random-weight draws per model, and block-bootstrap intervals. On the 1980–2025 test period, raising the number of features from 12 to 12,000 raises the timing strategy's Sharpe ratio from 0.03 to 0.22, while out-of-sample R^2 stays near -2.5% and no model beats the historical mean. A placebo test acts as the answer key. At the feature scale used by Kelly et al. ($\gamma = 2$), replacing the predictors with persistent noise removes the gain (Sharpe at most 0.05), and volatility-timed momentum explains only 8% of the strategy's variance. With wider kernels, under which nearby months look alike, the same noise reproduces much of the real model's Sharpe ratio (37–79%, depending on the kernel width and the standardization), as Nagel's argument predicts. Whether the "virtue of complexity" reflects learning therefore depends on a tuning choice, and in no setting does the strategy beat buy-and-hold.

1 Introduction

Predicting the stock market's monthly return is a classic test for statistical learning: the signal is weak, the sample is short, and the cost of fooling oneself is high. Goyal and Welch [Goyal and Welch, 2008] showed that popular predictors such as the dividend yield fail to beat the historical-mean forecast out of sample, and a recent update finds that most still fail [Goyal et al., 2024].

Kelly, Malamud and Zhou [Kelly et al., 2024] challenged this view. They argue that simple models *understate* predictability and that models with far more parameters than observations, ridge regression on thousands of random nonlinear features of about fifteen predictors, produce a profitable timing strategy even when trained on twelve months. Nagel [Nagel, 2025] offers a different explanation. When the number of features P greatly exceeds the window T, the ridge forecast becomes a similarity-weighted average of the T training returns. If similarity mostly reflects closeness in time, the forecast behaves like momentum, scaled down when predictors are volatile. The strategy then works because volatility-timed momentum happened to work, not because the model learns from the predictors.

The two explanations make different predictions. If the model learns from the predictors, replacing them with noise should remove its edge, and a transparent momentum rule should not explain its returns. If the gains are mechanical, they should survive noise that keeps the predictors' persistence.

Contributions. (i) A walk-forward replication at $T = 12$ in which every standardization and penalty uses past data only, averaged over 20 random-weight draws, with block-bootstrap intervals. (ii) Placebo tests with independent noise, persistent noise matched to each predictor's autocorrelation, and persistent noise whose volatility follows the real predictors. (iii) An attribution of strategy returns

Preprint.

to market, momentum and volatility-timed momentum. (iv) Evidence that the answer depends on the feature scale γ, which controls how similar nearby months look. (v) Post-publication (2021–2025) and trained-network comparisons.

2 Related work

Goyal and Welch [Goyal and Welch, 2008] established the historical mean as the benchmark, and Campbell and Thompson [Campbell and Thompson, 2008] introduced the out-of-sample R^2 we use; Gu, Kelly and Xiu [Gu et al., 2020] applied many machine-learning methods to return prediction. Random Fourier features [Rahimi and Recht, 2007] approximate a Gaussian kernel with a finite random basis. Work on double descent [Belkin et al., 2019] and ridgeless regression [Hastie et al., 2022] shows that test error can fall again once parameters outnumber observations; Kelly et al. [Kelly et al., 2024] bring this argument to finance. Time-series momentum [Moskowitz et al., 2012] and volatility-managed portfolios [Moreira and Muir, 2017] are well-documented return sources, and Nagel [Nagel, 2025] argues that random-feature ridge with short windows combines them mechanically.

3 Data and method

Data. We use the Goyal–Welch predictor file updated through December 2025, maintained by A. Goyal and posted publicly for research.[1] The target r_{t+1} is the S&P 500 total return minus the risk-free rate in month $t+1$. The $K = 15$ predictors known at the end of month t are the log dividend–price, dividend-yield, earnings–price and payout ratios, stock variance, book-to-market, net equity issuance, the Treasury-bill rate, the long-term yield and return, the term spread, the default yield and return spreads, inflation (lagged an extra month because CPI is published with a delay), and the current excess return. The sample has $N = 1{,}188$ monthly targets from January 1927 to December 2025 with no missing values; recent values match FRED and S&P Dow Jones Indices.

No look-ahead. Each predictor is standardized with its expanding mean and standard deviation up to month t (after a 36-month warm-up). Kelly et al. instead divide predictors by their expanding standard deviation without demeaning and scale training returns by their trailing 12-month volatility; we rerun the key results with that choice in Section 4. A forecast made at month t is trained only on pairs (z_s, r_{s+1}) with $s < t$. Penalties are chosen by out-of-sample R^2 on a validation period (targets 1950–1979, 360 months); the test period (1980–2025, 552 months) is scored once. Random K-fold cross-validation is not used because it would train on the future.

Models. The *historical mean* forecasts the average of all past excess returns. *OLS* and *lasso* use the 15 predictors with an intercept on an expanding window of at least 240 months. *Ridge* uses the 15 predictors on a rolling T-month window without an intercept. *Random-feature ridge* maps z_t to P features $\sin(\gamma\,\omega_i^\top z_t)$ and $\cos(\gamma\,\omega_i^\top z_t)$ with $\omega_i \sim N(0, I)$ and $\gamma = 2$, then fits ridge on a rolling window. With F the $T \times P$ training features scaled by $1/\sqrt{P}$, the forecast is computed in kernel form,

$$\hat{r}_{t+1} = f_t^\top F^\top \left(FF^\top + \lambda T\, I_T\right)^{-1} r_{\text{train}}, \tag{1}$$

which needs only a $T \times T$ solve. Equation (1) shows Nagel's point directly: the forecast is a weighted average of the T training returns, with weights set by how similar each training month's predictors are to today's. We vary P from 2 to 12,000; for each P we draw 20 independent weight sets and report the mean and range across draws, never the best draw. Penalty grids are $\lambda \in \{10^{-3}, \ldots, 10^{3}\}$ (ridge) and $\{10^{-5}, \ldots, 10^{-2}\}$ (lasso). The main window is $T = 12$, where Nagel's argument applies most strongly.

Evaluation. $R^2_{OOS} = 1 - \sum(r_{t+1} - \hat{r}_{t+1})^2 / \sum(r_{t+1} - \bar{r}_t)^2$ is measured against the historical mean $\bar{r}_t$. The timing strategy holds a position equal to the forecast and earns $\hat{r}_{t+1} r_{t+1}$; we report its annualized Sharpe ratio. Forecasts are serially correlated, so 95% intervals use a circular block bootstrap with 12-month blocks and 2,000 resamples; for random features, months are resampled jointly across all draws.

Placebos. We rerun random-feature ridge with the predictors replaced by noise, drawing new noise for each weight draw and tuning the penalty the same way. *Independent* noise has no persistence.

[1] `https://sites.google.com/view/agoyal145`. We cite the data and publish only derived results.

Table 1: Test period 1980–2025 (552 months), $T = 12$ unless noted. 95% block-bootstrap intervals in brackets. Random-feature results average 20 weight draws; the MLP averages 5 initializations. Buy-and-hold Sharpe ratio: 0.59.

Model	R^2_{OOS} (%)	Sharpe ratio
Historical mean	0	0.58 [0.27, 0.91]
OLS, expanding window	-10.61 [-17.77, -4.13]	0.07 [-0.16, 0.44]
Lasso, expanding window	-0.03 [-0.16, 0.08]	0.58 [0.27, 0.92]
Ridge, 15 predictors	-2.51 [-5.44, 0.13]	0.13 [-0.07, 0.41]
Random-feature ridge, $P = 12$	-2.84 [-5.95, -0.28]	0.03 [-0.05, 0.10]
Random-feature ridge, $P = 1{,}200$	-2.55 [-5.58, -0.01]	0.17 [-0.08, 0.41]
Random-feature ridge, $P = 12{,}000$	-2.49 [-5.52, 0.02]	0.22 [-0.13, 0.55]
MLP, $T = 12$ (extension)	-31.92 [-44.35, -19.83]	0.10 [-0.21, 0.35]
MLP, $T = 120$ (extension)	-24.41 [-37.57, -12.75]	0.20 [-0.13, 0.56]

Persistent noise is AR(1) with each predictor's own first-order autocorrelation (median 0.98), so nearby months look alike, as Nagel's mechanism requires, while the noise carries no information about returns. A model that learns from the predictors should lose its edge under both.

Attribution. Momentum is $\text{MOM}_t = \sum_{k=0}^{11} w_k r_{t-k}$ with linearly declining weights $w_k \propto 12 - k$; volatility-timed momentum is $\text{VTM}_t = \text{MOM}_t/\sigma_t$, where σ_t is the average 12-month volatility of changes in the standardized predictors. We regress each strategy's returns on the market and on the momentum strategies' returns, all scaled to 10% annual volatility, with Newey–West standard errors [Newey and West, 1987] (12 lags).

Extension. A one-hidden-layer network (64 tanh units, L2 penalty tuned on validation, 5 initializations) is trained on the same windows. Random-feature ridge is the same architecture with a frozen random first layer, so the comparison asks whether learning that layer helps.

4 Results

Baselines and replication. Table 1 reports the test period. Expanding-window OLS loses badly to the historical mean ($R^2_{OOS} = -10.6\%$), reproducing Goyal and Welch's finding. The validation period chooses lasso's largest penalty, which keeps at most two of the 15 coefficients, so lasso is nearly the historical mean. Ridge at $T = 12$ has $R^2_{OOS} = -2.5\%$ and a Sharpe ratio of 0.13.

Figure 1 shows the complexity curve. As P grows from 2 to 12,000, the Sharpe ratio rises from about 0 to 0.22 and the spread across draws narrows, while R^2_{OOS} stays near -2.5%. This is the qualitative pattern Kelly et al. report: better timing without better point forecasts. On their sample period (targets 1930–2020) the Sharpe ratio rises from 0.03 at $P = 12$ to 0.35 at $P = 12{,}000$ (buy-and-hold: 0.42), and R^2 against a zero forecast, their benchmark, is $+0.6\%$. That window overlaps our validation period, so these figures are not fully out of sample. On the test period every interval for the random-feature Sharpe ratio includes zero, and none reaches the historical mean or buy-and-hold.

Placebo. Figure 2(a) is the answer key. With independent or persistent noise in place of the predictors, the Sharpe ratio stays between -0.03 and 0.05 at every P. Two checks confirm this: fixing the penalty at the real model's value ($\lambda = 0.1$) instead of re-tuning gives Sharpe ratios between -0.05 and 0.05, and persistent noise whose volatility follows the real predictors' 12-month volatility, which keeps both ingredients of Nagel's mechanism, gives at most 0.05 [-0.07, 0.19]. At $\gamma = 2$ the gain from complexity needs the real predictors.

Attribution. Table 2 regresses strategy returns on the market and on volatility-timed momentum. For random-feature ridge at $P = 12{,}000$, the loading on VTM is positive and significant ($t = 3.2$), but VTM explains only 8% of the return variance, its forecasts correlate 0.30 with the VTM signal, and adding it lowers the alpha only from 1.43% to 1.07% a year at 10% volatility. Neither alpha is significant. By contrast, plain ridge at $T = 12$ is close to momentum: momentum explains 76% of its variance and removes its alpha.

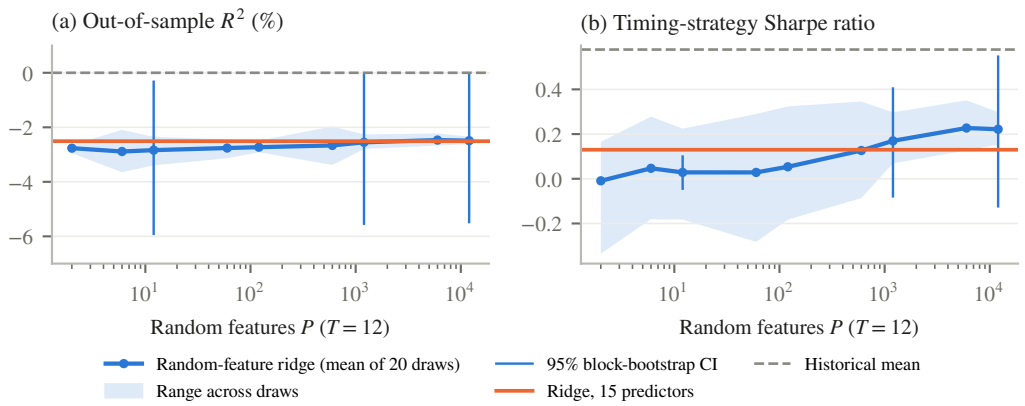


Figure 1: Complexity curve at $T = 12$, test period 1980–2025. (a) Out-of-sample R^2 against the historical mean. (b) Sharpe ratio of the timing strategy. Whiskers: 95% block-bootstrap intervals for the draw-averaged metric.

Table 2: Attribution, test period. Strategy returns regressed on the market (MKT) and on momentum (MOM) or volatility-timed momentum (VTM) strategy returns, all at 10% annual volatility. Alpha in % per year; Newey–West t-statistics in parentheses.

	Ridge, 15 predictors		Random features, $P = 12{,}000$	
	MKT + MOM	MKT + VTM	MKT + MOM	MKT + VTM
Alpha (MKT only)	1.19 (0.83)		1.43 (0.83)	
Alpha	−0.67 (−0.94)	0.03 (0.05)	1.06 (0.64)	1.07 (0.65)
Momentum loading	0.87 (8.4)	0.77 (5.7)	0.17 (2.8)	0.24 (3.2)
R^2	0.76	0.57	0.05	0.08

The answer depends on the feature scale. Random Fourier features approximate a Gaussian kernel whose width is $1/\gamma$. At $\gamma = 2$ with fifteen standardized predictors, the average similarity between consecutive months is only 0.07, so nearby months do not look alike and Nagel's mechanism has nothing to work with. Figure 2(b) repeats the placebo at wider kernels, choosing the penalty on validation as before. At $\gamma = 0.5$ (similarity 0.58) the real model earns 0.37 [0.10, 0.67] and persistent noise earns 0.25 [0.03, 0.47]; at $\gamma = 0.25$ (similarity 0.82) the figures are 0.36 and 0.29. Noise reproduces 13% of the real Sharpe ratio at $\gamma = 2$, 55% at $\gamma = 1$, 67% at $\gamma = 0.5$ and 79% at $\gamma = 0.25$. Where nearby months look alike, much of the edge is mechanical, as Nagel argues; where they do not, the mechanism switches off and the remaining edge is small and insignificant. With Kelly et al.'s standardization the $\gamma = 2$ results are unchanged (real 0.24 [−0.11, 0.54], persistent noise −0.03), and at $\gamma = 0.5$ noise reproduces 37% of the real Sharpe ratio (0.13 against 0.36) rather than 67%, so the size of the mechanical share depends on standardization even though its direction does not.

Which inputs matter. At $\gamma = 2$, the current excess return alone gives a Sharpe ratio of 0.12, the 14 economic predictors alone give 0.06, and the full set gives 0.22. Keeping the return but replacing the 14 predictors with persistent noise gives −0.04: noise in most dimensions swamps the kernel similarity.

Post-publication, longer windows and the trained network. On 2021–2025 (60 months) the strategy at $P = 12{,}000$ has a Sharpe ratio of −0.36 [−1.26, 0.74] against 0.77 for buy-and-hold; with 60 months the interval is too wide to conclude much. With $T = 60$ and $T = 120$ the Sharpe ratio at $P = 12{,}000$ is 0.22 and 0.24, so the $T = 12$ result is not special to the short window. The trained MLP at $T = 12$ overfits badly ($R^2_{OOS} = -31.9\%$) despite the validation period choosing the largest penalty. At $T = 120$ its Sharpe ratio is 0.20 [−0.13, 0.56], no better than random features over the same window (0.24), with a far worse R^2_{OOS} (−24.4%). Learning the first layer does not help here; with so little data, the frozen random layer acts as useful regularization.

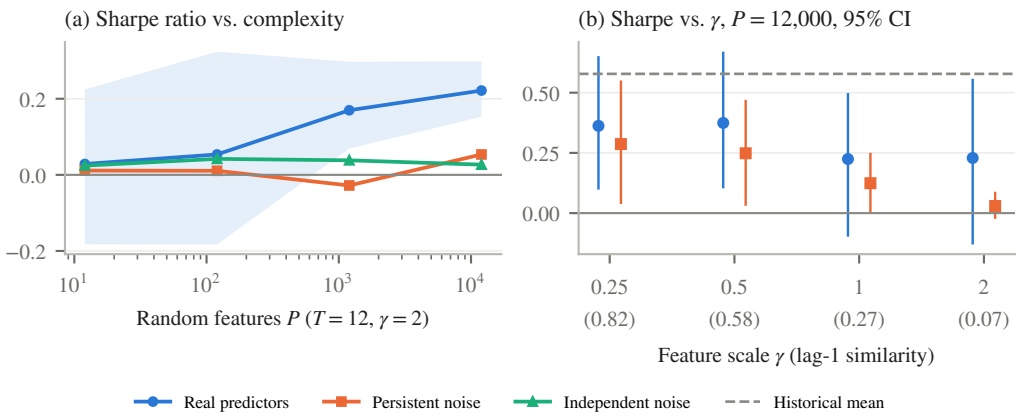


Figure 2: Placebo tests, test period. (a) Sharpe ratio against P at $\gamma = 2$; shading is the range across draws for real predictors. (b) Real predictors against persistent noise at $P = 12{,}000$ for four feature scales; the number under each γ is the average Gaussian-kernel similarity between consecutive months.

5 Limitations

Our main standardization differs from Kelly et al.'s; rerunning with theirs leaves the $\gamma = 2$ conclusions unchanged but lowers the mechanical share at wide kernels, and other unreported details may differ. We did not tune toward the original numbers. The momentum signals are transparent proxies for Nagel's mechanism, not his exact construction. Monthly returns carry little signal, so intervals are wide and the 2021–2025 check is weak. Strategy results ignore transaction costs and leverage limits, and cover one market index. Penalties are tuned on 1950–1979; the Kelly-period comparison overlaps that window.

6 Conclusion

At the feature scale used by Kelly et al., the gain from complexity in our replication needs the real predictors: noise does not reproduce it and volatility-timed momentum explains little of it. But the gain is small, never statistically distinguishable from zero, and below buy-and-hold. At wider kernels, persistent noise reproduces a large share of the edge (37–79%, depending on kernel width and standardization), which supports Nagel's mechanism. The lesson for practice is that a complexity curve alone cannot show that a model learns; a placebo with matched persistence can, and it should be reported alongside any claimed virtue of complexity.

Acknowledgments and Disclosure of Funding

AI use. We used an LLM assistant (Claude, Anthropic) for literature search, code generation, debugging and drafting of the manuscript. All experimental results were produced by the accompanying code and checked by the author. **Contributions.** Lifantao He is the sole author and carried out all parts of the project. **Data.** Goyal–Welch predictor data, courtesy of A. Goyal.

References

Mikhail Belkin, Daniel Hsu, Siyuan Ma, and Soumik Mandal. Reconciling modern machine-learning practice and the classical bias–variance trade-off. *Proceedings of the National Academy of Sciences*, 116(32):15849–15854, 2019.

John Y. Campbell and Samuel B. Thompson. Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4):1509–1531, 2008.

Amit Goyal and Ivo Welch. A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4):1455–1508, 2008.

Amit Goyal, Ivo Welch, and Athanasse Zafirov. A comprehensive 2022 look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 37(11):3490–3557, 2024.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.

Trevor Hastie, Andrea Montanari, Saharon Rosset, and Ryan J. Tibshirani. Surprises in high-dimensional ridgeless least squares interpolation. *Annals of Statistics*, 50(2):949–986, 2022.

Bryan Kelly, Semyon Malamud, and Kangying Zhou. The virtue of complexity in return prediction. *Journal of Finance*, 79(1):459–503, 2024.

Alan Moreira and Tyler Muir. Volatility-managed portfolios. *Journal of Finance*, 72(4):1611–1644, 2017.

Tobias J. Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen. Time series momentum. *Journal of Financial Economics*, 104(2):228–250, 2012.

Stefan Nagel. Seemingly virtuous complexity in return prediction. Working Paper 34104, National Bureau of Economic Research, 2025.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems*, volume 20, 2007.

A Additional results

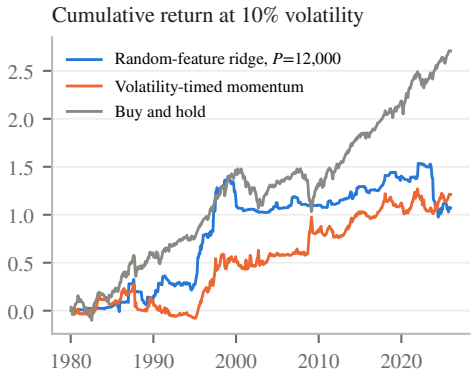


Figure 3: Cumulative returns over 1980–2025, each strategy scaled to 10% annual volatility (scaling uses full-period volatility and is for display only).

Table 3: Robustness, test period: Sharpe ratio (95% interval) of random-feature ridge at $P = 12{,}000$.

Setting	Sharpe ratio
$T = 12$, $\gamma = 2$ (main)	0.22 [−0.13, 0.55]
$T = 60$	0.22 [−0.08, 0.53]
$T = 120$	0.24 [−0.04, 0.52]
Persistent noise, penalty re-tuned	0.05 [−0.02, 0.12]
Persistent noise, $\lambda = 0.1$	−0.02 [−0.08, 0.05]
Volatility-matched persistent noise	0.05 [−0.07, 0.19]
Independent noise	0.03 [−0.03, 0.08]
Kelly et al. standardization	0.24 [−0.11, 0.54]
Kelly et al. standardization, persistent noise	−0.03 [−0.10, 0.04]
$\gamma = 0.5$, Kelly et al. standardization	0.36 [0.08, 0.63]
$\gamma = 0.5$, Kelly et al. std., persistent noise	0.13 [−0.10, 0.38]
Current return only	0.12 [−0.13, 0.39]
14 predictors, no current return	0.06 [−0.29, 0.38]

Implementation. Random features use $P/2$ sine and $P/2$ cosine features of the same $P/2$ random projections. Penalties chosen on validation: ridge $\lambda = 100$; random features $\lambda = 1$ for $P \leq 120$ and $\lambda = 0.1$ for $P \geq 600$ (except $\lambda = 10$ at $P = 2$); lasso $\alpha = 10^{-2}$. Seeds are fixed, and all results are produced by one notebook that runs end to end.